

4.3.3 Preventing, treating and curing diseases

The human body has defence systems to protect it from the pathogens that cause communicable diseases. However, these defences can be breached.

Vaccination helps to protect people from diseases that were once widespread. If the immune system fails, then antibiotics can be used to treat bacterial infections.

The increasing problem of antibiotic resistance (see [Evidence for evolution](#) (page 84)) means that research to develop new medicines has to continue. Clinical trials of new drugs have to be carefully planned and the results published so that claims can be subject to peer review and checked by other scientists replicating the investigations.

New technologies based on genetic modification and stem cells are making it possible to provide effective treatments for non-communicable diseases but, in many cases, these are still at an early stage of development. The development and application of new technologies in medicine can raise ethical issues.

4.3.3.1 Spread of communicable diseases

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how communicable diseases (caused by viruses, bacteria, protists and fungi) are spread in animals.	Harmful microorganisms (pathogens) that cause disease can spread: • through the air when people cough or sneeze • through food that is contaminated with bacteria • through drinking water that is contaminated with microorganisms • through contact with other people, or surfaces that infected people have touched • by animals that scratch, bite or draw blood.	WS 1.2 Apply the ideas in this section to the transmission of the common cold, flu, cholera, athlete's foot and malaria.

4.3.3.2 Human communicable diseases

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe a minimum of one common human infection, and sexually transmitted infections in humans, including HIV/AIDS. Explain how the spread of communicable diseases may be reduced or prevented in animals. This should include a minimum of one common human infection, and sexually transmitted infections in humans including HIV/AIDS.	Salmonella food poisoning is spread by bacteria ingested in food, or on food prepared in unhygienic conditions. Fever, abdominal cramps, vomiting and diarrhoea are caused by the bacteria and the toxins they secrete. Salmonella bacteria are killed by cooking and pasteurisation. In the UK, poultry are vaccinated against Salmonella to control the spread. Measles is a viral disease showing symptoms of fever and a red skin rash. Measles is a serious illness that can be fatal if complications arise. For this reason most young children are vaccinated against measles. The measles virus is spread by inhalation of droplets from sneezes and coughs. Gonorrhoea is a sexually transmitted disease (STD) with symptoms of a thick yellow or green discharge from the vagina or penis and pain on urinating. It is caused by a bacterium and was easily treated with the antibiotic penicillin until many resistant strains appeared. Gonorrhoea is spread by sexual contact. The spread can be controlled by treatment with antibiotics or the use of a barrier method of contraception such as a condom. HIV initially causes a 'flu like illness'. Unless successfully treated with antiretroviral drugs the virus attacks the body's immune cells. Late-stage HIV, or AIDS, occurs when the body's immune system is no longer able to deal with other infections or cancers. HIV is spread by sexual contact or exchange of body fluids such as blood.	WS 1.4 Explain applications of science to prevent the spread of diseases.

4.3.3.3 Defences against pathogens

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the non-specific defence systems of the human body against pathogens.	The human body defends itself against the entry of pathogens in the following ways: • the skin is a barrier and produces antimicrobial secretions • the nose catches particles • the trachea and bronchi secrete mucus that is moved by cilia • the stomach produces acid, which kills the majority of pathogens that enter via the mouth.	

4.3.3.4 The human immune system

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the role of the immune system of the human body in defence against disease.	If a pathogen enters the body the immune system tries to destroy the pathogen. White blood cells are an important part of the immune system. They help to defend against pathogens through: • phagocytosis • producing antibodies • producing antitoxins.	

4.3.3.5 Vaccination

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the use of vaccines in the prevention and treatment of disease.	Vaccination involves introducing small quantities of dead or inactive forms of a pathogen into the body to stimulate the white blood cells to produce antibodies. If the same pathogen re-enters the body the white blood cells respond quickly to produce antibodies, preventing infection. If a large proportion of the population is immune to a pathogen, the spread of the pathogen is very much reduced. Students do not need to know details of vaccination schedules and side effects associated with specific vaccines.	

4.3.3.6 Medicines

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the use of medicines in the prevention and treatment of disease.	Antibiotics, such as penicillin, are medicines that help to cure bacterial disease by killing infective bacteria inside the body. It is important that specific bacteria should be treated by specific antibiotics. The use of antibiotics has greatly reduced deaths from infectious bacterial diseases. However, the emergence of strains of bacteria resistant to antibiotics is becoming a serious threat. Antibiotics cannot kill viral pathogens. Painkillers and other medicines are used to treat the symptoms of disease. They do not kill pathogens.	This topic links with Variation and evolution (page 82).
Explain that many useful materials are formulations of mixtures.	Most medicines are mixtures. They are formulations made by mixing the ingredients in carefully measured quantities to ensure that the product has the required properties. One or more of the ingredients may be the drug, such as aspirin, but other ingredients make it easier or more pleasant for a patient to take the drug in solution or as a capsule or tablet.	

4.3.3.7 Testing new drugs

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the process of discovery and development of potential new medicines, including preclinical and clinical testing.	When new medical drugs are devised, they have to be extensively tested and trialled before being used. Drugs are tested in a series of stages to find out if they are safe and effective. New drugs are extensively tested for toxicity, efficacy and dose: • in the laboratory, using cells, tissues and live animals • then in clinical trials involving healthy volunteers and patients. Very low doses of the drug are given at the start of the clinical trial. If the drug is found to be safe, further clinical trials are carried out to find the optimum dose for the drug. In double-blind trials, some patients are given a placebo. Patients are allocated randomly to groups so that neither the doctors nor the patients know who has received a placebo and who has received the drug until the trial is complete.	WS 1.6 Explain that the results of testing and trials, like the findings of all scientific research, are published only after evaluation by peer review.

4.3.3.8 Genetic modification

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain some of the possible benefits and risks, including practical and ethical considerations, of using gene technology in modern medicine.	New medical products have been produced by genetically modifying bacteria. Insulin for the treatment of Type 1 diabetes is produced by cultivating genetically modified bacteria. Sheep and goats have been genetically modified to produce chemicals in their milk that can be used to treat disease. In one example the milk produced contains a protein needed to treat patients with cystic fibrosis. Research is also exploring the possibility of providing tissues needed for transplants from animals that have been genetically modified so that the tissues are not rejected by the human immune system.	WS 1.3 Evaluate gene technologies, taking into account benefits, risks, and the ethical issues raised by the use of animals in medical research. This topic links with Variation and evolution (page 82).

4.3.3.9 Stem cells

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Discuss potential benefits and risks associated with the use of stem cells in medicine.	One medical use of stem cells is well established: this is the use of stem cells from bone marrow in transplants to provide a supply of new blood cells for the person receiving the transplant. This is used to treat leukaemia. Stem cells for research may be based on: • stem cells from embryos that are a few days old • adult stem cells from selected parts of the body such as bone marrow • fetal stem cells taken from blood in the umbilical cord. Embryonic stem cells can develop into any of the many types of cells in the body. Adult stem cells can only give rise to the types of cells found in the tissues that the adult stem cells come from. Most medical uses of stem cells are still experimental. Treatments based on stem cells are being investigated for treating diseases such as: • heart disease – using the patient's own stem cells from bone marrow • Type 1 diabetes – using embryo or fetal stem cells. The properties of stem cells are not fully understood. Scientists do not yet know how their differentiation is controlled. This means that there is a fear that their ability to proliferate could lead to cancer when they are transplanted into a patient.	WS 1.3 Give a simple ethical argument about the rights and wrongs of the uses of stem cells. Evaluate possible uses of stem cells taking into account benefits, risks and the ethical issues raised by sources of the cells.

4.3.3.10 Interactions between different types of disease

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the interactions between different types of disease.	Different types of disease may interact. Some examples include: • defects in the immune system mean that an individual is more likely to suffer from infectious diseases • viruses living in cells can be the trigger for cancers • immune reactions initially caused by a pathogen can trigger allergies such as skin rashes and asthma • severe physical ill health can lead to depression and other mental illness.	